Quantifying the Gaps: A Systematic Taxonomy of Bias and Imbalance in Multilingual AI Evaluation

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Abstract

Multilingual AI now mediates access to information and services for billions of speakers across the world's 7,000+ languages, yet the evaluation infrastructure that certifies its progress is fragmented and poorly understood. We introduce a systematic, capability-aligned taxonomy of multilingual evaluation, individually annotating 96 resources spanning 139 languages (65 evaluation benchmarks, 2018–2025; 31 training datasets, 2005–2025) along five empirically motivated dimensions: data origin, linguistic resource tier, task class, cultural specificity, and inference paradigm. To our knowledge this is the first audit to treat cultural specificity and inference paradigm as first-class axes and to examine the training-data supply jointly with benchmarks, complementing score-level sweeps of 2,000+ benchmarks. A quantitative meta-analysis converts scattered intuitions about the field's biases into measurable structural facts: only 6.2% of evaluation benchmarks center low-resource languages, leaving roughly 1.2 billion speakers effectively unmeasured; 26.2% of benchmarks are built on translated data, importing performance-distorting artifacts; 55.4% are culturally non-specific; and question answering plus syntax account for nearly half of all benchmarks. Cross-referencing dimensions with reported results surfaces consequential dynamics, including the consistent advantage of Translate-Test inference over direct multilingual inference in low-resource settings—evidence that current leaderboards overstate native multilingual competence. We publicly release the annotated corpus and classification criteria, and distill the analysis into an actionable roadmap for funding, building, and reporting the evaluation infrastructure that globally equitable AI requires.

1 Introduction

Language technology increasingly determines who can use AI at all. More than 7,000 languages are spoken today, and an estimated 88% of them—the languages of roughly 1.2 billion people—have effectively no language-technology resources at all (Joshi et al. 2020). Yet the capabilities of multilingual models are certified by an evaluation ecosystem concentrated on a few dozen languages. Benchmarks are not passive measurement instruments: they direct research attention, gate model releases, and shape funding and deployment decisions (Blodgett et al. 2020; Raji et al. 2021). When evaluation under-represents a language community, the consequences compound: models ship with untested failure modes, apparent parity on translated tests masks real-world breakage, and the absence of measurement is quietly read as absence of demand. The costs are already visible beyond AI research: health-system roadmaps identify weak local-language performance as a leading barrier to LLM adoption in low- and middle-income countries (Chen et al. 2025), and UNESCO guidance asks education systems to validate linguistic appropriateness before deploying generative AI (Miao and Holmes 2023). The equity of multilingual evaluation is therefore not a methodological footnote but a precondition for AI that serves a global population.

The community has responded to this challenge with resources rather than with understanding. Benchmarks and corpora have proliferated—over 2,000 multilingual benchmarks appeared between 2021 and 2024 alone (Wu et al. 2025)—into a fragmented landscape in which comparing results, identifying systemic weaknesses, and directing new effort grow steadily harder. The structure of this ecosystem—as distinct from its volume—has not been systematically measured: the field's primary challenge is no longer creating more resources but constructing a conceptual map of the ones that already exist.

This paper provides that map. Rather than another narrative survey, we perform a deconstruction and reorganization of the multilingual evaluation landscape. Its 96 resources—65 evaluation benchmarks and 31 training datasets spanning 139 languages—are individually annotated along five dimensions selected to probe the primary failure modes of multilingual AI: *data origin* (native vs. translated), *linguistic resource tier*, *task class*, *cultural specificity*, and *inference paradigm*. The result is a structured, queryable representation of the field's evaluation infrastructure, released in full.

The taxonomy turns qualitative unease into quantitative diagnosis. Low-resource languages constitute 6.2% of evaluation benchmarks, while 70.8% of benchmarks target predominantly high-resource languages. More than a quarter (26.2%) of benchmarks are built on translated rather than natively sourced data, importing well-documented "translationese" artifacts into the measurement instrument itself. A majority (55.4%) are culturally non-specific, and two task

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families—question answering and syntactic evaluation—absorb 49.2% of all benchmarks. Cross-referencing these annotations with reported results exposes dynamics invisible to any single resource, including the consistent finding that routing low-resource inputs through English ("Translate-Test") outperforms direct multilingual inference, a measure of how far native multilingual competence lags behind what leaderboards imply.

Our contributions are:

- **A five-dimensional, capability-aligned taxonomy** of multilingual evaluation resources, with explicit classification criteria designed to expose structural bias—a reusable protocol for auditing evaluation ecosystems beyond multilingual NLP.
- **An annotated corpus of 96 resources** (65 benchmarks, 31 training datasets, 139 languages), publicly released with per-resource structured summaries.[1]
- **A quantitative meta-analysis** that measures the field's imbalances in resource tier, data origin, task coverage, and cultural grounding at the level of verified structural annotations, and organizes reported results into cross-dimensional performance dynamics.
- **An evidence-based roadmap** translating each measured gap into concrete guidance for benchmark builders, funders, model auditors, and the language communities the evaluations are meant to serve.

2 Related Work

Prior syntheses of multilingual NLP are predominantly *model-centric*: surveys of multilingual LLMs, cross-lingual transfer, and social bias catalog methods and behaviors, and a recent survey organizes culturally aware NLP around a taxonomy of cultural elements (Liu, Gurevych, and Korhonen 2025). None provides a structural, per-resource accounting of the evaluation infrastructure itself.

Closest to our work, Wu et al. (2025) analyze over 2,000 multilingual benchmarks at the level of aggregate scores, documenting English over-representation and showing that native benchmarks track human judgments far better than their translated counterparts (§4.2). Their breadth-first, score-level evidence is complementary to ours: we contribute a depth-first *structural* annotation of five manually assessed axes, including cultural specificity and inference paradigm, which no automated sweep captures. We also audit the training-data supply side jointly with benchmarks, enabling cell-level gap analysis (e.g., low-resource $\times$ reasoning) that aggregate scores cannot support. Our smaller corpus is a design choice, not a concession: every resource passed an explicit three-part screen and was annotated from primary documentation (§3), so each cell of the analysis is individually auditable. Concurrent work audits 51 multilingual benchmarks under coverage, representativeness, and rigor pillars and reaches a congruent diagnosis of thin, translation-heavy coverage (Uppadhyay et al. 2026); it annotates neither resource tier nor inference paradigm and does not examine the training-data supply side.

Our method descends from the documentation tradition of datasheets (Gebru et al. 2021) and data statements (Bender and Friedman 2018), and from resource audits of linguistic diversity (Joshi et al. 2020). We operationalize these ideas at field scale: rather than documenting one artifact, we impose a common documentation schema on an entire research area and analyze the result as data—the methodology of established structural audits that have annotated 100+ safety datasets (Röttger et al. 2025), 445 benchmarks for construct validity (Bean et al. 2025), and 205 web-crawled corpora for quality (Kreutzer et al. 2022).

3 Corpus and Taxonomy

3.1 Corpus Construction

We conducted a targeted sweep of the primary venues of AI research—the ACM Digital Library, arXiv, the ACL Anthology, and Hugging Face Datasets—for multilingual and cross-lingual evaluation resources (benchmarks published 2018–2025; training corpora 2005–2025). Candidates passed a three-part screen: (i) public availability of the dataset or a detailed benchmark description; (ii) explicit multilingual or cross-lingual scope; and (iii) clear, reproducible evaluation results or metrics. After screening and deduplication, 96 resources were retained: 65 evaluation benchmarks and 31 training datasets covering 139 unique languages. A small number of influential monolingual resources were retained only where they anchor an explicitly multilingual lineage: BLiMP's minimal-pair template, for instance, is the design ancestor of syntactic benchmarks in Chinese, Turkish, Russian, and Dutch, and of a 101-language successor (Warstadt et al. 2020; Liu et al. 2026; Başar et al. 2025; Taktasheva et al. 2024; Suijkerbuijk et al. 2025; Jumelet et al. 2026).

3.2 Five Dimensions of Evaluation

Each resource is annotated along five axes chosen to probe a distinct failure mode of multilingual AI, aligning the taxonomy with the capabilities deployment depends on rather than with dataset genres.

D1: Data origin (*native* | *translated*). Confronts the "translationese" problem by quantifying reliance on machine- or human-translated content versus natively sourced data.

D2: Linguistic resource tier (*high* | *medium* | *low*). Following established resource-level classifications (Joshi et al. 2020), maps the chasm between resource-rich and resource-poor languages.

D3: Task class (13 classes: QA, syntax, reasoning, multimodal, NLU, summarization, math, RAG, code, generation, long-context, embeddings, other). Reveals which capabilities the community actually measures.

D4: Cultural specificity (*universal* | *culturally anchored* | *region-specific*). The most consequential axis: distinguishes benchmarks that are genuinely culture-independent from

[1] The complete annotated corpus, classification criteria, and per-resource structured summaries accompany this submission as supplementary material; upon publication the corpus will be archived with a DOI and mirrored on a public dataset hub for community extension.

Dimension	Category	Bench. (65)	Data. (31)
Origin	Native	48 (73.8%)	23 (74.2%)
	Translated	17 (26.2%)	8 (25.8%)
Tier	High	46 (70.8%)	24 (77.4%)
	Medium	15 (23.1%)	5 (16.1%)
	Low	4 (6.2%)	2 (6.5%)
Cultural grounding	Universal	36 (55.4%)	22 (71.0%)
	Cult. anchored	17 (26.2%)	6 (19.4%)
	Region-specific	12 (18.5%)	3 (9.7%)
Purpose (datasets)	Pre-training	—	20 (64.5%)
	Translation	—	8 (25.8%)
	Instruction FT	—	3 (9.7%)

Table 1: Structural profile of the corpus: 65 evaluation benchmarks (2018–2025) and 31 training datasets (2005–2025) spanning 139 unique languages.

#	Language	n	#	Language	n
1	English	37	11	Korean	26
2	Chinese	35	12	Italian	25
3	Spanish	31	13	Vietnamese	24
4	German	30	14	Thai	23
5	French	30	15	Indonesian	23
6	Russian	30	16	Turkish	18
7	Arabic	28	17	Dutch	18
8	Japanese	28	18	Czech	15
9	Hindi	27	19	Polish	14
10	Portuguese	27	20	Greek	13

Table 2: Benchmark coverage of the twenty most-covered languages (n = number of the 65 benchmarks including the language). Each of the remaining 119 covered languages appears in at most 13 benchmarks; more than 6,800 living languages appear in none.

those that merely strip cultural context, quantifying Anglo-centric and Western-centric defaults (Singh et al. 2025b).

D5: Inference paradigm. Captures the nature of the reasoning demanded (retrieval-style vs. multi-hop) and the evaluation regime (direct multilingual inference vs. Translate-Test) (Conneau et al. 2018), probing the brittleness of measured performance.

3.3 Annotation and Meta-Analysis

Each resource was annotated on every applicable axis from its primary paper, dataset card, and repository: all five axes apply to evaluation benchmarks, while for training datasets task class is recorded as training purpose (pre-training, translation, or instruction fine-tuning) and the inference-paradigm axis does not apply. Mixed cases follow explicit decision rules: a resource combining native and translated subsets is classified by its majority composition, and resource tier reflects the tier of the languages a benchmark is *designed to evaluate*, not incidental coverage. The classification criteria and complete annotations are released with this paper, so every cell of our analysis can be audited and corrected. We then synthesized the annotations in a quantitative meta-analysis, cross-referencing dimensions to identify macro-level trends—treating the field's evaluation infrastructure itself as the object of study.

4 Findings: Structural Imbalances and Their Dynamics

Table 1 and Figure 1 summarize the corpus. Four imbalances dominate the benchmark landscape (§4.1–§4.4), the training-data supply mirrors them (§4.5), and their interactions produce performance dynamics that no single resource reveals (§4.6).

4.1 A Pervasive Skew Toward High-Resource Languages

Only 4 of 65 benchmarks (6.2%)—and 2 of 31 datasets (6.5%)—center low-resource languages, while 70.8% of benchmarks and 77.4% of datasets target predominantly high-resource ones (Table 1). The language-level view is starker (Table 2): English appears in 37 of 65 benchmarks, and coverage collapses past a short head—every language outside the twenty most-covered appears in at most 13 benchmarks. Compressed to a single inequality index, benchmark coverage across the world's $\sim$7,000 living languages carries a Gini coefficient between 0.982 and 0.994 regardless of how the unobserved tail is allocated (derivation in the supplement). Nominal breadth also flatters the field: multi-way parallel suites such as BELEBELE's 122 language variants (Bandarkar et al. 2024) multiply coverage by translating a single task—breadth without native depth.

The corpus documents what this skew costs. Where benchmarks do reach beyond the head, measured capability drops sharply: MMLU-ProX reports declines of up to 24.3% for low-resource languages across 36 evaluated LLMs (Xuan et al. 2025); code-specialized models such as WizardCoder and DeepSeek-Coder collapse on low-resource languages in mHumanEval's 204-language extension (Raihan, Anastasopoulos, and Zampieri 2025); and in multilingual long-context evaluation, gaps *widen* as context grows, with English ranking only sixth—behind Polish—at long contexts (Kim et al. 2025). These failure modes are invisible to the other 93.8% of benchmarks, which seldom test the languages where breakage concentrates.

4.2 Translated Foundations Distort Measurement

More than a quarter of benchmarks (26.2%) and datasets (25.8%) are built on translated data—including the canonical cross-lingual evaluation suite: XNLI, XQuAD, MLQA, and PAWS-X are all translation-constructed (Conneau et al. 2018; Artetxe, Ruder, and Yogatama 2020; Lewis et al. 2020; Yang et al. 2019). Translation was a rational bootstrapping strategy, but the corpus now contains direct evidence of a double distortion: translated suites not only import translationese artifacts into the measurement instrument but also inflate absolute scores and change model rankings. Global-MMLU finds that 28% of MMLU questions require culturally sensitive knowledge and 84.9% of its geographic questions concern North America or Europe; re-evaluating 14 mod-

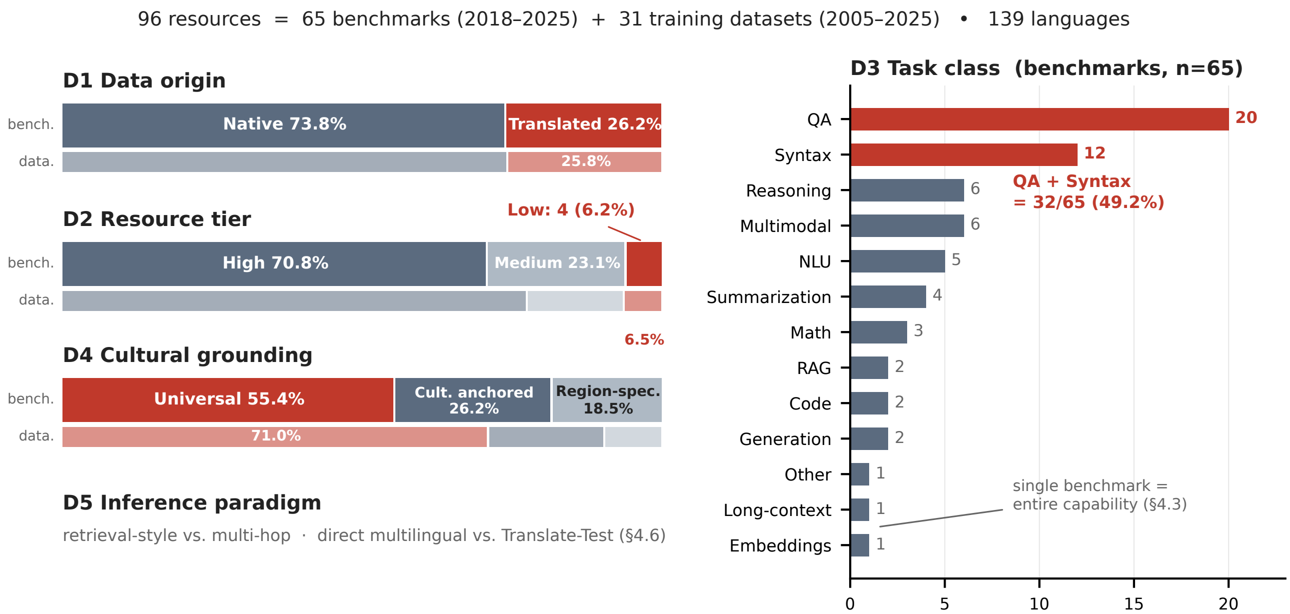

Figure 1: The five-dimensional taxonomy applied to the corpus (rows D1, D2, D4—thick bars: 65 evaluation benchmarks; thin bars: 31 training datasets). Highlighted segments mark the measured imbalances: 26.2% of benchmarks are built on translated data (D1); only 4 of 65 (6.2%) center low-resource languages (D2); QA and syntax absorb 49.2% of all benchmarks, while long-context, embeddings, and open-ended generation are each represented by one or two benchmarks (D3); and 55.4% are culturally non-specific (D4). The training-data supply shows the same or stronger skews: 25.8% translated, 6.5% low-resource, and 71.0% culturally universal—exceeding the benchmarks' 55.4%. D5 (inference paradigm) is analyzed in §4.6.

els on culturally sensitive versus culturally agnostic subsets *reorders the model leaderboard* (Singh et al. 2025b). Benchmarks built from native sources depress inflated scores: GPT-4 scores 72.5% on natively sourced ArabicMMLU—below its score on translated Arabic MMLU (Koto et al. 2024)—and 59.95% on Korean-native KMMLU, far below the ~80% pass mark of the source exams (Son et al. 2025b), with the same pattern on China-specific CMMLU questions (Li et al. 2024), while on Turkish-native TurkishMMLU most open models fall below 60% (Yüksel et al. 2024). On MultiLoKo's locally sourced questions in 31 languages, the best of 11 evaluated LLMs averages 34.4% (Hupkes and Bogoychev 2025). Score-level evidence converges: in Chinese, the native CMMLU correlates with human judgments at 0.68 versus 0.47–0.49 for translated MMLU variants (Wu et al. 2025). Where translation is unavoidable, rigorous pipelines with expert post-editing (e.g., IndicMMLU-Pro's back-translation and 13-reviewer verification) are the defensible middle path (Sankalp et al. 2025)—but they remain a proxy for, not a substitute for, native evaluation.

4.3 A Task Monoculture

Question answering (20 of 65; 30.8%) and syntactic evaluation (12 of 65; 18.5%) together absorb 49.2% of all benchmarks (Figure 1, D3), and the syntax count is partly template replication—BLiMP-style minimal pairs re-instantiated per language, up to 101 languages at once (Warstadt et al. 2020; Jumelet et al. 2026). Meanwhile the capabilities on which deployed systems depend are nearly unmeasured outside English: retrieval-augmented generation, code generation, and open-ended generation account for 3.1% each; long-context understanding and embeddings for 1.5% each—for the latter two, a *single* benchmark stands for the entire capability (Kim et al. 2025; Enevoldsen et al. 2025). The supply side mirrors the imbalance (§4.5): task-targeted multilingual training data is equally scarce. Operationally, the field has defined "multilingual capability" as answering questions and judging grammaticality; on that definition, models can score well while remaining untested on the tasks users will actually give them.

4.4 Cultural Grounding Is the Neglected Dimension

A majority of benchmarks (55.4%) and 71.0% of datasets are "universal"—designed to be culture-neutral, which in practice usually means cultural context has been stripped or inherited from Anglophone sources. Only 26.2% of benchmarks are culturally anchored and 18.5% region-specific. Language, however, is a cultural construct, and culturally agnostic evaluation certifies models that fail in situ. On the culturally curated multimodal ALM-bench, GPT-4o falls from 78.8% overall to 50.8% on Amharic (Vayani et al. 2025); on ECLEKTIC's cross-lingual knowledge-transfer test, the best model achieves only 41.6% overall success (65% transfer score) (Goldman et al. 2025); and SeaEval documents inconsistent answers to semantically equivalent queries across languages (Wang et al. 2024). The countermovement—native school and professional exams (ArabicMMLU, CMMLU, KMMLU, TurkishMMLU), the African-led, expert-translated IrokoBench (Adelani et al. 2025), and participatory regional hubs such as SEACrowd (Lovenia et al. 2024)—consistently reveals gaps that translated suites miss, which is precisely the argument for funding more of it.

4.5 The Supply Side Feeds the Same Gaps

Auditing training datasets alongside benchmarks shows the two sides of the ecosystem reinforcing each other. Pre-training corpora dominate the supply (64.5% of the 31 datasets), and the largest of them inherit the web's language distribution even at trillion-token scale (Penedo et al. 2024; Nguyen et al. 2024; Kudugunta et al. 2023; Laurençon et al. 2022). Translation and alignment corpora account for another 25.8%. Multilingual *instruction* data—the ingredient that turns a pre-trained model into a usable assistant—amounts to three datasets (9.7%) (Muennighoff et al. 2023; Li et al. 2023). Of these, only the participatory Aya Dataset, human-curated by fluent speakers of 65 languages, provides natively authored instructions at scale; its companion Aya Collection reaches 513M instances across 114 languages largely by templating and translating existing datasets (Singh et al. 2024). The pattern parallels or exceeds the benchmark side's skews: 74.2% native, 77.4% high-resource, and 71.0% culturally universal versus the benchmarks' 55.4% (Table 1). Evaluation gaps and data gaps are the same gap seen from two directions, and closing one without the other leaves models that either cannot learn a capability or cannot prove they have it.

4.6 Cross-Dimensional Dynamics

Cross-referencing axes exposes three performance dynamics with direct practical consequences, and the temporal record adds a fourth.

The Translate-Test paradox. Across independent low-resource evaluations, routing inputs through English outperforms direct multilingual inference: Indic-QA finds Translate-Test significantly better than direct inference, especially for low-resource Indic languages (Singh et al. 2025a); XNLI's translation-based baselines remain the strongest (Conneau et al. 2018); and on AmericasNLI's ten Indigenous languages, zero-shot cross-lingual transfer collapses to 30–50% accuracy while translation-based approaches lead (Ebrahimi et al. 2022). The seemingly inefficient pipeline winning is an indictment: much of what leaderboards certify as multilingual competence is English competence accessed through translation—the prevailing paradigms measure the bridge, not the destination—and single-paradigm reporting hides the gap.

Complexity is where multilinguality breaks. The inference-paradigm axis shows measured competence degrading as tasks move from retrieval toward composition. Even in an English monolingual RAG setting, the best frontier model reaches only 62.4%; with genuinely multilingual retrieval it manages at most 57.6%—against human upper bounds of 85–92%—and on cross-lingual RAG with monolingual retrieval it falls to 55.5% (other evaluated models: 31–41%), frequently answering in the wrong language outright (Liu et al. 2025). Overall success on cross-lingual knowledge transfer tops out at 41.6% (§4.4). And multilingual long-context aggregation tasks remain essentially unsolved, with under 1% accuracy on the hardest variant (Kim et al. 2025). Benchmarks confined to retrieval-style QA—the plurality of the corpus—systematically overstate how robust "multilingual" capability is.

Scale is not a language policy. Larger models lift averages without closing gaps: BenchMAX finds English–non-English gaps persist across model sizes (Huang et al. 2025); P-MMEval shows measured transfer depends on the benchmark's language of origin (Zhang et al. 2025); test-time scaling gains are English-centric (+20 points on AIME vs. +1.9 averaged elsewhere) (Son et al. 2025a); and multilingual chain-of-thought emerges only at extreme scale (Shi et al. 2023). Waiting for scale to deliver equity is not a strategy.

The surge is steerable. Benchmark creation accelerated sharply—38 of 65 benchmarks (58.5%) date from 2024–2025 alone (Figure 2)—and the same window produced a rising corrective thread: community-built resources, often natively sourced and culturally grounded (Singh et al. 2024; Lovenia et al. 2024; Adelani et al. 2025; Enevoldsen et al. 2025). The imbalances we quantify are not destiny; they are a snapshot of momentum that the community can redirect while the paradigm is still forming.

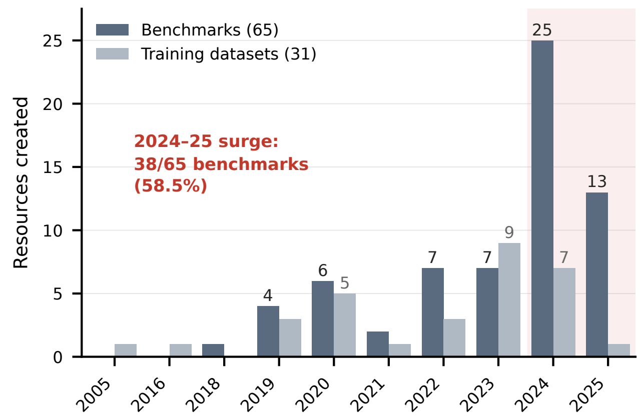

Figure 2: Resources created per year: 65 evaluation benchmarks (2018–2025) and 31 training datasets (2005–2025; empty intermediate years omitted). More than half of all benchmarks (38 of 65; 58.5%) appeared in 2024–2025 alone—a field young enough that its imbalances reflect current momentum rather than settled legacy (§4.6). The surge is evaluation-only: through 2023 benchmark and dataset production ran near parity (27 vs. 23), but in 2024–2025 benchmarks outpaced new training datasets 38 to 8—nearly five-fold (Fisher's exact $p = 0.004$)—certifying capabilities faster than the data to build them is being supplied (§4.5).

5 From Diagnosis to Action

The taxonomy is built to be used, not admired. We identify its users and translate each finding into a recommendation.

Benchmark builders can treat the five axes as a coverage card at design time and target empty cells directly: low-resource × reasoning, low-resource × long-context, region-specific × anything beyond QA. **Funders and program officers** gain quantified evidence for allocating scarce annotation budgets toward native data creation rather than further translation of saturated high-resource suites. **Model developers and auditors** can require evaluation-provenance disclosure—the native/translated share, cultural-specificity mix, and inference paradigm behind a claimed "multilingual" score—before accepting deployment claims for a region. **Language communities and regional consortia** can use the corpus to locate what exists for their languages and the taxonomy to name what is missing, converting absence into a fundable agenda—following the participatory models of Aya, SEACrowd, and IrokoBench (Singh et al. 2024; Lovenia et al. 2024; Adelani et al. 2025).

The stakes are practical, not academic. Multilingual models are already procured for education, public-health information, and government services in regions whose languages the certifying evaluations never natively tested. A system that scores well on translated, culture-stripped suites can still misinform a farmer in Amharic or a patient in Quechua while its leaderboard entry says otherwise (§4.2, §4.4, §4.6). The risk is already quantified: on expert-annotated health questions, GPT-3.5's correctness drops by up to 38.6% in Hindi relative to English (Jin et al. 2024)—and Hindi is among the ten best-covered languages in our corpus (Table 2), so measured harm is plausibly a lower bound for the languages no benchmark reaches. Roadmaps for LLM adoption in low- and middle-income health systems likewise name suboptimal local performance as a primary barrier (Chen et al. 2025). Governments are already moving: UNESCO's global guidance on generative AI in education directs member states to validate linguistic appropriateness before deployment (Miao and Holmes 2023), and the European Commission funds sovereign multilingual LLMs on the explicit ground that existing models under-serve even EU languages (European Commission 2025). As language requirements enter AI procurement and national AI strategies, *evaluation provenance*—what a "multilingual" claim was actually measured on—belongs in those requirements. The taxonomy supplies the vocabulary and the released annotations supply the evidence base for writing them.

The findings support five recommendations. **R1:** Prioritize natively sourced, culturally grounded benchmark creation in low-resource tiers, where 6.2% benchmark coverage meets the largest measured performance drops (§4.1, §4.4) and roughly 1.2 billion speakers remain effectively unmeasured (§1). **R2:** Treat translated benchmarks as transitional scaffolding: label data origin explicitly, and report native-source results wherever both exist (§4.2). **R3:** Diversify the task portfolio beyond QA and syntax toward reasoning, generation, retrieval, and long-context capabilities that deployment depends on (§4.3). **R4:** Report both direct multilingual and Translate-Test performance; the gap between them is itself a diagnostic of native competence (§4.6). **R5:** Adopt the five axes as a reporting standard for new evaluation resources—a datasheet for benchmarks—so the next generation of resources is born documented (§3.2).

The complete annotations and classification criteria are released in machine-readable form to support integration with dataset hubs and leaderboard metadata, and we invite the community to refine and extend the corpus as the field evolves.

6 Limitations

Our corpus is a curated snapshot of a fast-moving field, not an exhaustive census: breadth is deliberately traded for per-resource verification, and complementary large-scale sweeps (Wu et al. 2025) cover volume we do not. The convergence is itself informative: this structural annotation, the score-level sweep, and a concurrent coverage-and-representativeness audit (Uppadhyay et al. 2026) reach the same diagnosis of thin, translation-heavy coverage by methodologically independent routes, so the central findings do not rest on any single sampling or annotation choice. The five-axis annotation involves judgment; labels follow the explicit decision rules above and were checked against each resource's primary documentation, but we do not report a formal inter-annotator agreement statistic. As mitigation, the classification criteria and complete annotations are released so that every label can be audited and corrected by the community—the open-audit practice established by prior resource audits that likewise rely on single-pass structural labels (Kreutzer et al. 2022; Lovenia et al. 2024). The taxonomy records the *existence* and structure of resources, not their adoption or item-level quality. Finally, the performance dynamics in §4 synthesize results reported by the primary studies under heterogeneous protocols; they are observational evidence, not controlled re-evaluations.

7 Conclusion

We introduced a five-dimensional taxonomy of multilingual AI evaluation, applied it to 96 resources, and quantified the field's structural imbalances: a 6.2% share for low-resource languages, more than a quarter of benchmarks built on translation, nearly half of all benchmarks devoted to two task families, and a majority of resources stripped of cultural grounding. These are measurements of an infrastructure that currently certifies "multilingual" AI for a world it does not represent. The audit protocol itself—a screened corpus, per-resource structural axes keyed to documented failure modes, and released auditable labels—transfers to other evaluation ecosystems, such as speech and multimodal benchmarks, where the same questions of origin, tier, and cultural grounding remain unasked. By reframing evaluation as foundational infrastructure—mapped, documented, and steerable—this work gives benchmark builders, funders, auditors, and language communities a shared evidence base for building AI that is not just multilingual in theory but globally reliable and culturally inclusive in practice.

Ethical Statement

All 96 annotated resources are publicly documented research artifacts; our corpus contains no personal data and required no human-subjects involvement. Two risks of this work itself merit disclosure. First, taxonomy labels could be misread as quality judgments: a "universal" label means a benchmark strips or abstracts cultural context, not that it is validated for all cultures, and annotation errors could misdirect funding—which is why every label is released for public audit and correction. Second, the annotators' own linguistic and cultural vantage points shape judgment-based axes such as cultural specificity; the same public-audit mechanism is the intended check on both risks. The work's aim is distributive: to shift evaluation investment toward the language communities that current infrastructure leaves unmeasured.

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